

MOSKVA SHEREMETYEVO, Russian Federation

WMO: 275155

Lat: 55.973N

Lon: 37.415E

Elev: 190

StdP: 99.07

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.9	(c) -20.2	(d) -27.0	(e) 0.3	(f) -23.6	(g) -23.8	(h) 0.4	(i) -20.2	(j) 10.0	(k) -3.6	(l) 9.1	(m) -4.2	(n) 2.6	(o) 70	(p) 0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.9	(c) 30.0	(d) 19.2	(e) 28.0	(f) 18.5	(g) 26.0	(h) 18.1	(i) 20.7	(j) 26.1	(k) 19.7	(l) 25.2	(m) 18.9	(n) 24.4	(o) 4.1	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.1	22.8	18.0	13.2	22.1	17.0	12.4	21.0	60.3	25.9	57.2	25.0	54.5	24.5	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.2	8.3	7.4	-26.5	31.6	3.8	2.4	-29.3	33.3	-31.5	34.7	-33.6	36.0	-36.4	37.8
			-26.7	22.1	3.8	1.1	-29.5	22.9	-31.7	23.5	-33.8	24.1	-36.6	24.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-7.5	-7.2	-1.9	6.2	12.8	16.4	19.3	17.2	11.7	5.3	-0.8	-5.2	(6)
(7)		DBStd	10.71	6.57	6.64	4.82	4.81	4.85	3.96	3.47	3.69	3.97	4.31	5.27	6.38	(7)
(8)		HDD10.0	2529	542	481	369	133	27	2	0	0	27	153	324	471	(8)
(9)		HDD18.3	4772	800	714	628	363	181	83	30	65	202	403	574	729	(9)
(10)		CDD10.0	928	0	0	0	20	113	195	288	225	77	9	1	0	(10)
(11)		CDD18.3	128	0	0	0	0	9	26	59	32	2	0	0	0	(11)
(12)		CDH23.3	1181	0	0	0	3	96	255	519	291	17	0	0	0	(12)
(13)		CDH26.7	323	0	0	0	0	19	61	158	84	1	0	0	0	(13)
(14)	Wind	WSAvg	3.9	4.2	4.2	4.2	3.8	3.8	3.7	3.5	3.5	3.5	4.0	4.2	4.4	(14)
(15)	Precipitation	PrecAvg	665	47	39	36	34	55	71	79	74	66	66	52	45	(15)
(16)		PrecMax	878	84	78	73	57	116	118	167	145	185	134	87	80	(16)
(17)		PrecMin	437	16	8	14	12	17	9	13	13	12	16	3	14	(17)
(18)		PrecStd	108	17	14	16	13	25	26	40	37	33	36	19	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	3.2	12.9	23.2	29.0	30.8	33.8	32.9	25.2	18.8	11.2	7.8	(19)
(20)			MCWB	3.3	1.8	7.3	13.1	17.7	18.6	19.5	19.3	16.7	13.0	9.7	7.4	(20)
(21)		2%	DB	2.1	2.3	8.9	20.1	26.0	28.2	30.1	28.9	22.9	15.8	8.8	5.0	(21)
(22)			MCWB	1.6	1.3	4.3	11.5	16.7	18.3	19.5	19.0	15.9	12.3	7.9	4.4	(22)
(23)		5%	DB	1.2	1.9	6.0	17.2	23.2	26.2	28.1	26.2	20.2	13.2	7.0	3.1	(23)
(24)			MCWB	0.7	1.1	2.9	10.1	14.9	17.9	18.9	18.4	14.8	10.6	6.3	2.5	(24)
(25)		10%	DB	0.2	1.0	4.1	14.9	21.2	24.1	26.1	24.2	18.1	11.2	5.2	1.9	(25)
(26)			MCWB	-0.3	0.4	1.9	8.7	14.0	16.8	18.6	17.4	14.0	9.3	4.6	1.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	2.6	7.4	14.0	18.9	21.3	22.6	21.2	18.2	14.2	10.3	7.0	(27)
(28)			MCDWB	3.4	2.9	11.9	22.2	26.5	27.4	28.8	27.5	22.8	16.4	11.4	7.2	(28)
(29)		2%	WB	1.5	1.6	4.9	12.2	17.6	19.8	21.1	20.0	16.6	12.6	7.7	4.2	(29)
(30)			MCDWB	1.7	2.0	7.6	18.4	24.3	25.4	26.0	25.7	21.0	15.2	8.1	4.6	(30)
(31)		5%	WB	0.6	0.9	3.2	10.8	16.4	18.7	20.0	19.1	15.5	11.2	6.2	2.5	(31)
(32)			MCDWB	0.9	1.3	5.0	16.1	21.8	24.4	25.2	25.0	19.2	13.3	6.5	2.8	(32)
(33)		10%	WB	-0.2	0.3	2.2	9.6	15.0	17.6	19.3	18.1	14.4	9.9	4.5	1.0	(33)
(34)			MCDWB	0.2	0.8	3.8	14.1	19.5	22.6	24.5	23.2	17.8	11.0	5.0	1.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.0	6.2	7.3	9.6	10.6	10.1	9.9	9.8	8.4	5.9	4.2	4.3	(35)
(36)			MCDBR	3.6	4.4	9.5	13.6	13.3	13.0	13.0	13.0	11.9	8.9	5.5	3.5	(36)
(37)			MCWBR	3.7	4.2	6.2	6.9	6.0	5.5	4.7	5.2	6.0	5.9	4.9	3.6	(37)
(38)		5% WB	MCDWR	3.6	3.9	7.6	12.3	11.6	11.4	10.7	11.2	10.3	8.0	4.6	3.5	(38)
(39)			MCWBR	3.9	4.1	5.5	6.8	5.7	5.4	4.8	5.2	5.7	5.7	4.5	3.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.292	0.335	0.349	0.421	0.402	0.387	0.433	0.438	0.388	0.358	0.323	0.287	(40)	
(41)		taud	2.115	2.068	2.115	2.115	2.299	2.362	2.253	2.222	2.343	2.354	2.228	2.111	(41)	
(42)		Ebn at Noon	585	696	798	791	839	856	803	769	760	685	551	488	(42)	
(43)		Edh at Noon	61	95	116	135	120	114	125	121	94	73	56	48	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.57	1.42	2.96	4.00	5.24	5.47	5.37	4.31	2.74	1.25	0.48	0.30	(44)	
(45)		RadStd	0.11	0.20	0.29	0.36	0.48	0.53	0.58	0.45	0.36	0.22	0.08	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.83	N/A	N/A	N/A	N/A	N/A	-180	-253	N/A	N/A
(47) Regional (10 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	-147	-219	+110	N/A

Nomenclature: See separate page